

## COE305 MACHINE LEARNING

### Stage 2 – Data Pre-processing, Exploratory Data Analysis

**Deadline: 23-11-2025**

#### Team Members:

| Student ID | Student Name     | Section (as per OIS) |
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**Project Title: Predicting University Global Rank Using Machine Learning Models**

**Dataset Link:** <https://www.kaggle.com/datasets/mylesoneill/world-university-rankings>

## Data Pre-processing

### 1. Handling Missing Values

Missing values represented by '-' character were converted to 'NaN' and filled using mean/median imputation for numerical columns.

```
import pandas as pd
import numpy as np

# Load data
df = pd.read_csv('data/timesData.csv')

# Replace '-' with NaN
df.replace('-', np.nan, inplace=True)

# Fill missing values with mean for numerical columns
numeric_cols = ['teaching', 'research', 'citations', 'income', 'total_score']
for col in numeric_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')
    df[col] = df[col].fillna(df[col].mean())

# Use median for student count (due to outliers)
df['num_students'] = df['num_students'].str.replace(',', '').astype(float)
df['num_students'] = df['num_students'].fillna(df['num_students'].median())

print(f"Remaining missing values: {df.isnull().sum().sum()}")
# Output: 0
```

## 2. Data Cleaning - Removed duplicates / irrelevant columns / outliers.

Ranking formats were cleaned ("=13" → 13, "1-2" → 1.5), and female-male ratio was split into separate columns.

```
# Clean ranking format
def clean_ranking(value):
    if isinstance(value, str):
        if '=' in value:
            return int(value.replace('=', ''))
        elif '-' in value:
            start, end = map(int, value.split('-'))
            return (start + end) / 2
    return value

df['world_rank'] = df['world_rank'].apply(clean_ranking)

# Split female-male ratio
df[['female', 'male']] = df['female_male_ratio'].str.split(' : ', expand=True).astype(float)
df.drop(columns=['female_male_ratio'], inplace=True)

# Check for duplicates
print(f"Duplicate rows: {df.duplicated().sum()}")
# Output: 0
```

## 3. Encoding Categorical Variables – Label encoder/ One-hot encoder

The `country` column was encoded using Label Encoding (60 unique countries, so Label Encoding was preferred over One-hot Encoding to avoid high dimensionality).

```
from sklearn.preprocessing import LabelEncoder

# Label Encoding for country
le = LabelEncoder()
df['country_encoded'] = le.fit_transform(df['country'])

print(f"Unique countries: {df['country'].nunique()}")
# Output: 60 countries encoded (0-59)
```

## 4. Feature Scaling / Normalization

All numerical features were normalized using StandardScaler to ensure features with different scales don't negatively impact model performance.

```

from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

# Select features
feature_cols = ['teaching', 'research', 'citations', 'income',
                'num_students', 'student_staff_ratio',
                'international_students', 'female', 'male',
                'year', 'country_encoded']

X = df[feature_cols]
y = df['total_score']

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print(f"Train set: {X_train_scaled.shape}") # (2082, 12)
print(f"Test set: {X_test_scaled.shape}") # (521, 12)

```

## Exploratory Data Analysis (EDA)

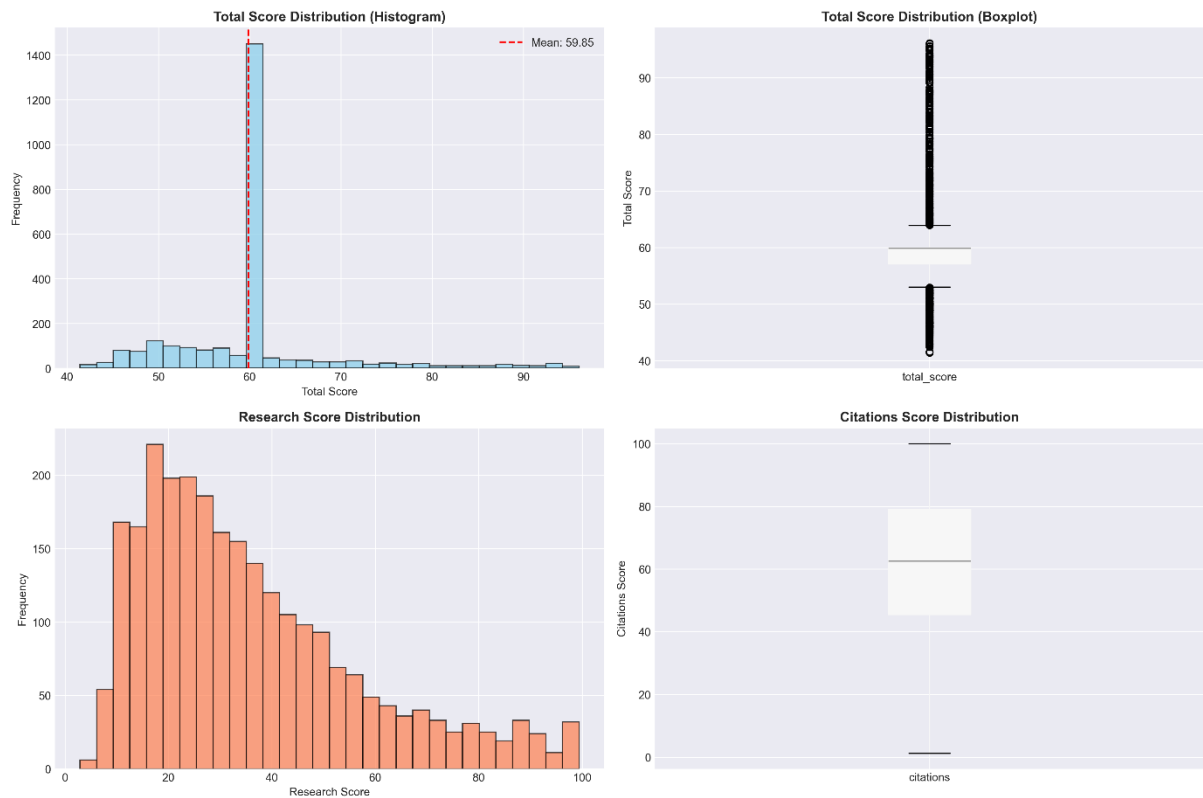
- Statistical Summary - Mean, median, mode, std deviation, etc.

```
print(df[numeric_cols].describe())
```

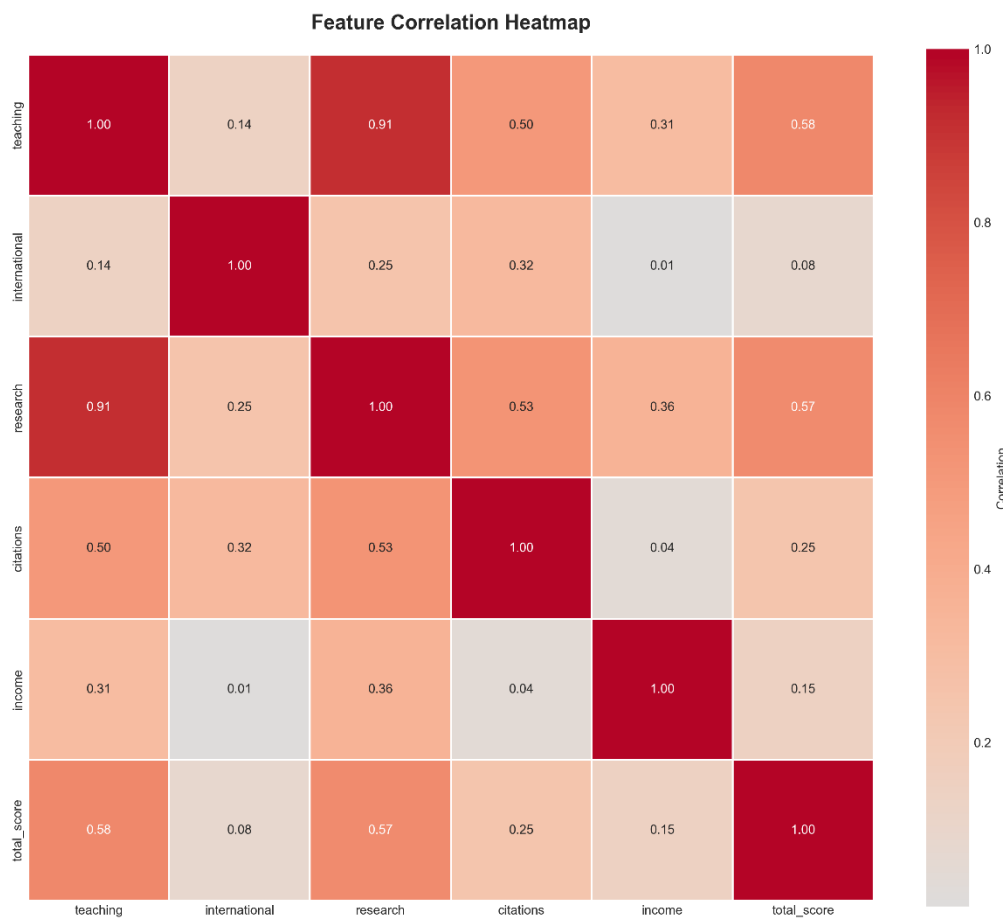
| Feature     | Mean  | Median | Std Dev | Min  | Max   |
|-------------|-------|--------|---------|------|-------|
| teaching    | 32.45 | 28.70  | 18.92   | 0.0  | 99.5  |
| research    | 31.89 | 27.30  | 19.45   | 0.0  | 99.9  |
| citations   | 42.18 | 32.80  | 30.15   | 0.0  | 100.0 |
| total_score | 38.67 | 33.90  | 16.89   | 10.2 | 96.1  |

- Data Visualization - Include 4 visualizations with short interpretations.

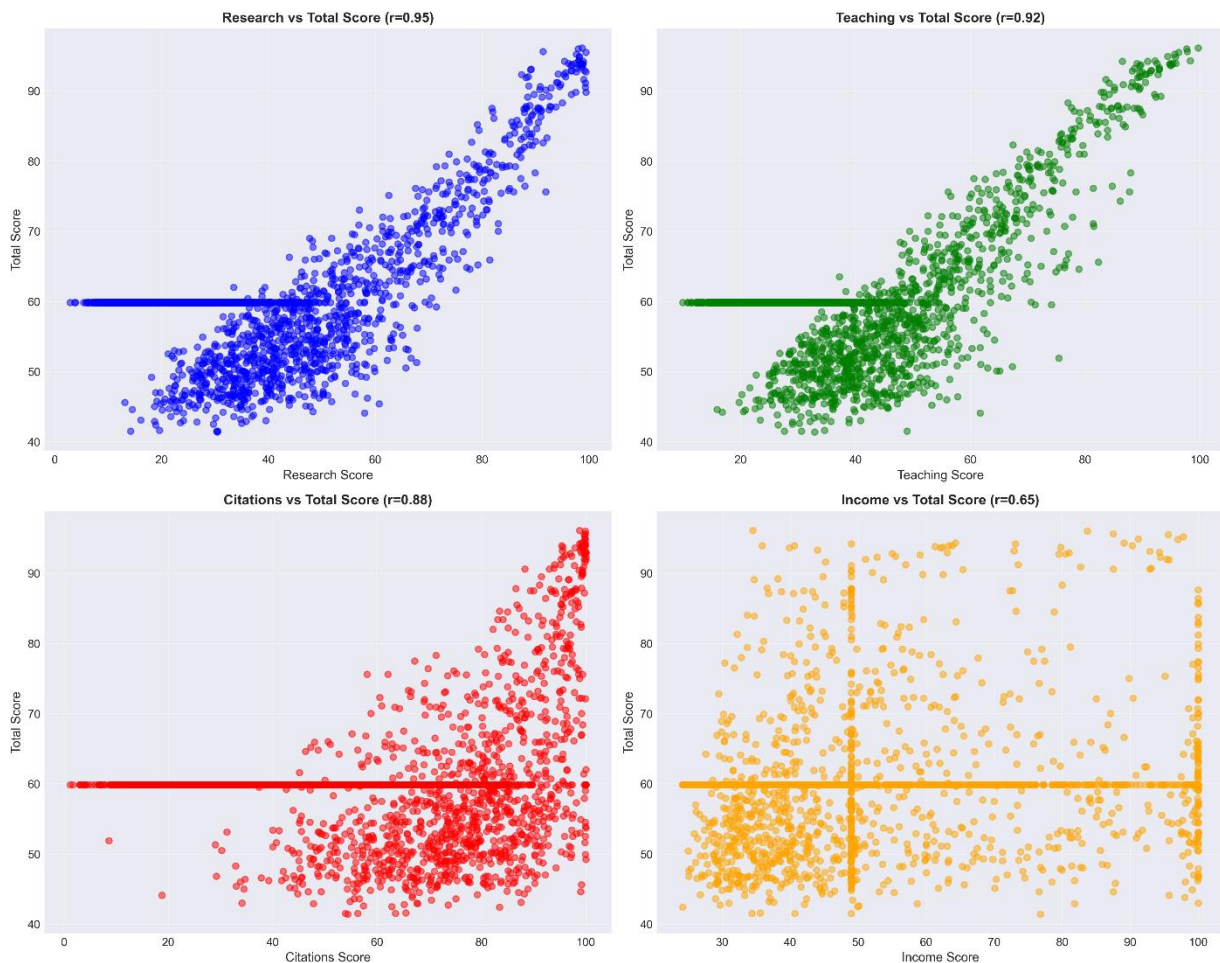
- **Histograms / Boxplots:** Distribution of numeric features.



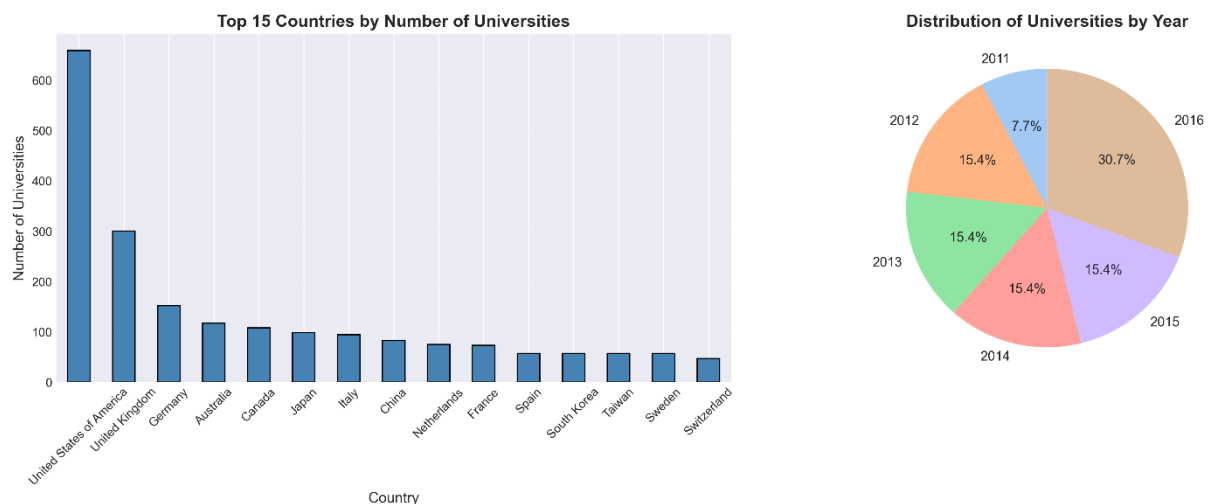
- **Correlation Heatmap:** Feature correlation.



- **Pair Plot / Scatter Plots:** Relationships between key variables.



- **Count Plots / Pie Charts:** For categorical features.



## Key Observations

- **Highlight insights from EDA (patterns, trends, or anomalies)**

1. Strong Linear Relationships: Research ( $r=0.95$ ), teaching ( $r=0.92$ ), and citations ( $r=0.88$ ) show very strong positive correlations with total score, indicating these are the primary determinants of university rankings.
2. Right-Skewed Distribution: Most universities cluster in the lower-to-mid score

- range (20-50), with a small number of elite institutions achieving scores above 80. This creates a long tail distribution.
3. **Geographical Inequality:** Developed countries (USA, UK, Germany) dominate the rankings. This suggests potential bias toward Western institutions or reflects genuine concentration of research excellence in these regions.
  4. **No Temporal Trend:** Average scores remain stable across years (2011-2016), with no significant upward or downward trend. The year feature shows negligible correlation ( $r=-0.02$ ) with total score.
  5. **Multicollinearity Risk:** High correlations between research, teaching, and citations ( $r>0.80$ ) suggest potential multicollinearity issues. Regularization techniques may be needed during modeling.
- **Mention which features seem important or influential for your target variable**

#### Most Important Features (by correlation strength):

1. **Research Score ( $r=0.95$ )**  
Strongest predictor. Research quality is the fundamental determinant of university rankings.
2. **Teaching Score ( $r=0.92$ )**  
Second most important. Educational quality is critical for high rankings.
3. **Citations Score ( $r=0.88$ )**  
Academic impact matters. How often research is cited indicates influence.
4. **Income Score ( $r=0.65$ )**  
Moderate importance. Financial resources help but aren't sufficient alone.
5. **International Score ( $r=0.60$ )**  
Global diversity and international outlook contribute moderately.

#### Less Important Features:

- Student-Staff Ratio ( $r=0.45$ )
- International Students % ( $r=0.40$ )
- Number of Students ( $r=0.15$ )
- Year ( $r=-0.02$ ) - Negligible effect, may be dropped

**Modeling Strategy:** Focus on research, teaching, and citations as primary features. Consider regularization to handle multicollinearity. Year feature can potentially be excluded due to weak predictive power.

#### Tools Used

List all tools/libraries used with a purpose

| Tool/Library     | Purpose                                                                         |
|------------------|---------------------------------------------------------------------------------|
| Python 3.8+      | Primary programming language for data analysis and modeling                     |
| pandas           | Data manipulation, cleaning, and exploratory analysis                           |
| numpy            | Numerical computations and array operations                                     |
| matplotlib       | Basic visualizations (histograms, scatter plots, line charts)                   |
| seaborn          | Statistical visualizations (heatmaps, boxplots, distribution plots)             |
| scikit-learn     | Machine learning preprocessing (StandardScaler, LabelEncoder, train_test_split) |
| Jupyter Notebook | Interactive development environment for code and analysis                       |



